OpenSource Connections

Agentic Relevance Tuning: Faster, Easier, and Higher Quality Search at Scale

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About Me 👋

🔍 **Search Consultant at OSC**

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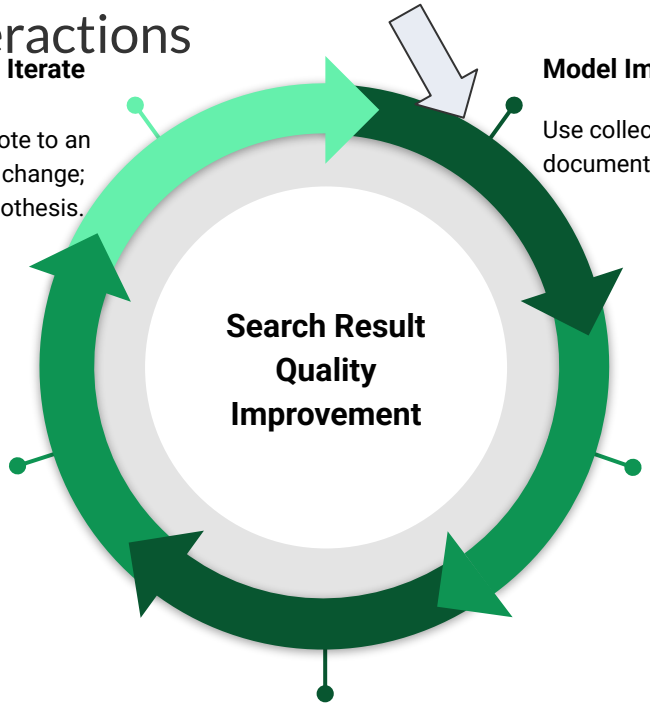
📚 **O'Reilly Author**

Relevance "Death Valley"

- HIPPO[1]: "Search results for X are terrible"
- Dig into logs & tracking data
- Analyze scoring, similarity calculations, complex query structures, boost functions, …
- Ultimately: You come up with a fix for X
- But: Fix for X breaks 10 other queries

[1] HIPPO: **Hi**ghest **P**aid **P**erson's **O**pinion



Model Implicit Relevance Judgments

Use collected user signals to infer document relevance for each query.

Measure Search Quality

Compute search quality metrics (e.g., NDCG, MRR, Recall, Precision).

Generate Hypotheses for Improvement

Identify weaknesses in ranking, retrieval, or query understanding.

Experiment & Validate Improvements

Measure search quality again using the same metrics to determine improvement.

Evaluate & Iterate

If search quality improves, promote to an online test & deploy the change; otherwise, refine the hypothesis.

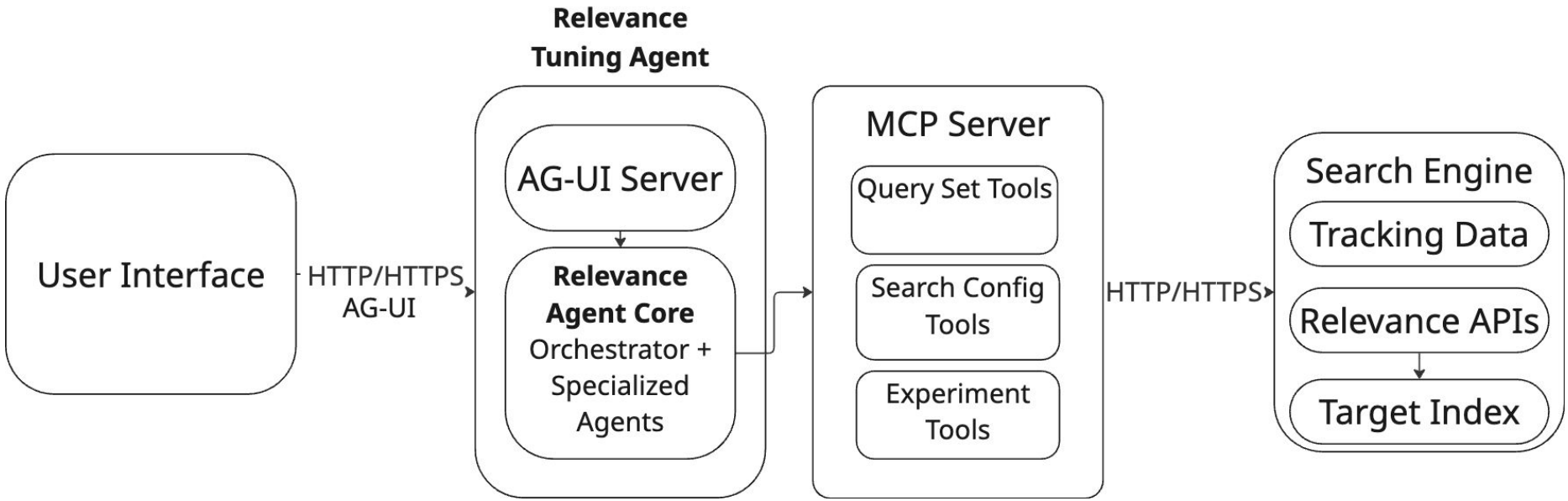
Enter: Relevance Tuning Agent

Why do everything by hand?

- Model the search result quality improvement process as a multi-agent system.
- Specialized sub-agents have access to tools and data sources:

The **User Behavior Analysis Agent**: Analyzes UBI data to identify relevance issues.
The **Hypothesis Generator Agent**: Interprets results, creates several hypotheses how to improve the identified issue
The **Evaluator Agent**: Automatically runs these changes against your offline evaluation sets in your Search Relevance Workbench.

Relevance Tuning Multi-Agent Architecture



Will everyone working in search relevance adjacent roles be out of jobs soon? No! Relevance tuning with agents is meant to …

- … **reduce the "Time-to-Fix"** from days or weeks to hours.
- … **lower the entry bar for relevance tuning** especially for newer engineers.
- … **increase confidence for relevance tuning changes** by providing automation around the experimentation process.

Lessons Learned

- Multi-agent architecture: Functionally looks like a good decision but we need to investigate in latency improvements
- Tools:
  - Project start: ~50 custom tools & 0 tools via MCP
  - Now: 1 custom tool left, ~20 contributed to MCP server, removed the rest
- Security is an interesting challenge touching upon multiple components of the whole stack
- Relying on standards can make your life easier: MCP & AG-UI meant easier integrability moving from a “greenfield PoC” to an MVP
- The “agent space” is moving extremely fast.
- Evals are critical.

Additional Discussion and Questions

Get in touch! Connect with me on LinkedIn or reach out to dwrigley@o19s.com

Book a 30 min chat about the relevance tuning agent - I'd love to hear your thoughts!

Resources

- OpenSearch Agent Server: Give it try and let me hear your feedback!
- Haystack Conference: Week of 4 May, tickets on sale now at haystackconf.com

Images: Nano Banana (slide 5), imgflip.com (slide 3 & slide 7)